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### Bowed runners and corresponding valve assemblies for paravalvular leak protection

#### Abstract

A prosthetic heart valve for replacing a native valve includes a stent extending between a proximal end and a distal end and including a plurality of struts forming cells, the stent having a collapsed condition and an expanded condition. At least one runner is coupled a cell, the at least one runner being configured to transition from a first configuration to a second configuration when the stent moves from the collapsed condition to the expanded condition, the at least one runner projecting radially outwardly from the cell in the second configuration. A valve assembly is disposed within the stent, the valve assembly including a plurality of leaflets, a cuff at least partially disposed on a luminal surface of the stent, and a covering material disposed on an abluminal surface of the stent and covering the at least one runner in the second configuration.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation of U.S. patent application Ser. No. 17/185,310 filed Feb. 25, 2021, which is a continuation of U.S. patent application Ser. No. 16/299,969 filed Mar. 12, 2019 (now U.S. Pat. No. 10,952,847), which is a continuation of U.S. patent application Ser. No. 15/921,106, filed Mar. 14, 2018 (now U.S. Pat. No. 10,271,946), which is a continuation of U.S. patent application Ser. No. 15/118,991 (now U.S. Pat. No. 9,949,825), filed Aug. 15, 2016, which is a national phase entry under 35 U.S.C. § 371 of International Application No. PCT/US2015/015533 filed Feb. 12, 2015, published in English, which claims priority from U.S. Provisional Patent Application No. 61/941,012, filed Feb. 18, 2014, the disclosures of which are incorporated herein by reference.

## BACKGROUND OF THE INVENTION

(1) The present disclosure relates in general to heart valve replacement and, in particular, to collapsible prosthetic heart valves. More particularly, the present disclosure relates to devices and methods for positioning and sealing collapsible prosthetic heart valves within a native valve annulus.

(2) Prosthetic heart valves that are collapsible to a relatively small circumferential size can be delivered into a patient less invasively than valves that are not collapsible. For example, a collapsible valve may be delivered into a patient via a tube-like delivery apparatus such as a catheter, a trocar, a laparoscopic instrument, or the like. This collapsibility can avoid the need for a more invasive procedure such as full open-chest, open-heart surgery.

(3) Collapsible prosthetic heart valves typically take the form of a valve structure mounted on a stent. There are two common types of stents on which the valve structures are mounted: a self-expanding stent or a balloon-expandable stent. To place such valves into a delivery apparatus and ultimately into a patient, the valve must first be collapsed or crimped to reduce its circumferential size.

(4) When a collapsed prosthetic valve has reached the desired implant site in the patient (e.g., at or

near the annulus of the patient's heart valve that is to be replaced by the prosthetic valve), the prosthetic valve can be deployed or released from the delivery apparatus and re-expanded to full operating size. For balloon-expandable valves, this generally involves releasing the entire valve, and then expanding a balloon positioned within the valve stent. For self-expanding valves, on the other hand, the stent automatically expands as the sheath covering the valve is withdrawn.

## SUMMARY OF THE INVENTION

(5) In some embodiments, a prosthetic heart valve for replacing a native valve includes a stent extending between a proximal end and a distal end and including a plurality of struts forming cells, the stent having a collapsed condition and an expanded condition. At least one runner is coupled to a cell, the at least one runner being configured to transition from a first configuration to a second configuration when the stent moves from the collapsed condition to the expanded condition, the at least one runner projecting radially outwardly from the cell in the second configuration. A valve assembly is disposed within the stent, the valve assembly including a plurality of leaflets, a cuff at least partially disposed on a luminal surface of the stent and a covering material disposed on an abluminal surface of the stent and covering the at least one runner in the second configuration.

(6) In some embodiments, a prosthetic heart valve for replacing a native valve includes a stent extending between a proximal end and a distal end and including a plurality of struts forming cells and a plurality of runners, the stent having a collapsed condition and an expanded condition, the struts defining a first diameter and the runners defining a second diameter, the second diameter being greater than the first diameter. A valve assembly is disposed within the stent, the valve assembly including a plurality of leaflets and a cuff at least partially disposed on a luminal surface of the stent and partially disposed on an abluminal surface of the stent to cover the runner.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments in accordance with the present disclosure will now be described with reference to the appended drawings. It is to be appreciated that these drawings depict only some embodiments and are therefore not to be considered limiting of its scope.

(2) FIG. 1 is a side elevational view of a conventional prosthetic heart valve;

(3) FIG. 2 is a highly schematic cross-sectional view taken along line A-A of FIG. 1 and showing the prosthetic heart valve disposed within a native valve annulus;

(4) FIG. 3A is a highly schematic side view of one embodiment of a heart valve having bowed runners intended to fill irregularities between the heart valve and the native valve annulus;

(5) FIG. 3B is a developed view of the stent of the heart valve of FIG. 3A in the collapsed configuration;

(6) FIGS. 3C and 3D are enlarged highly schematic partial views of a stent cell having a runner in the collapsed configuration and bowed configuration, respectively;

(7) FIGS. 4A-D are highly schematic side views of one method of delivering and deploying the heart valve of FIG. 3A within the native valve annulus;

(8) FIGS. 5A-10B are enlarged highly schematic partial side views of several variations of collapsed stent cells having runners and their respective shapes in the bowed configuration;

(9) FIGS. 11A-C are enlarged highly schematic side views of heart valves having bowed runners at various longitudinal positions; and

(10) FIG. 12 is a highly schematic cross-sectional view of a heart valve having bowed runners disposed within a native valve annulus;

(11) FIGS. 13-15 are highly schematic developed views of portions of heart valves including cuffs having folding flaps for covering runners in a first row;

(12) FIGS. 16A and 16B are highly schematic partial views of another method of folding a cuff

over runners in a first row;

(13) FIG. 17 is a highly schematic developed view of a portion of a heart valve having a cuff and individual panels attached over runners in a first row;

(14) FIGS. 18-20 are highly schematic developed views of portions of heart valves including cuffs having folding flaps attached over runners in a second row;

(15) FIG. 21 is a highly schematic developed view of a portion of a heart valve including a cuff and a covering disposed over runners in a second row; and

(16) FIGS. 22A-24 are highly schematic developed views of portions of heart valves including cuffs having various folding flaps and panels for covering runners in a third row.

#### DETAILED DESCRIPTION

(17) Despite various improvements that have been made to the collapsible prosthetic heart valve delivery process, conventional devices suffer from some shortcomings. For example, with conventional self-expanding valves, the clinical success of the valve is dependent on accurate deployment and anchoring. Inaccurate deployment and anchoring of the valve increases risks, such as those associated with valve migration, which may cause complications due to the obstruction of the left ventricular outflow tract. Inaccurate deployment and anchoring may also result in the leakage of blood between the implanted heart valve and the native valve annulus, commonly referred to as paravalvular leakage (also known as “perivalvular leakage”). In aortic valves, this leakage enables blood to flow from the aorta back into the left ventricle, reducing cardiac efficiency and putting a greater strain on the heart muscle. Additionally, calcification of the aortic valve may affect performance and the interaction between the implanted valve and the calcified tissue is believed to be relevant to leakage, as will be outlined below.

(18) Moreover, anatomical variations from one patient to another may cause a fully deployed heart valve to function improperly, requiring removal of the valve from the patient or performing an additional valve-in-valve procedure. Removing a fully deployed heart valve increases the length of the procedure as well as risks. Thus, methods and devices are desirable that would reduce the need to remove a prosthetic heart valve from a patient. Methods and devices are also desirable that would reduce the likelihood of paravalvular leakage due to gaps between the implanted heart valve and patient tissue.

(19) There therefore is a need for further improvements to the devices, systems, and methods for positioning and sealing collapsible prosthetic heart valves. Specifically, there is a need for further improvements to the devices, systems, and methods for accurately implanting a prosthetic heart valve. Among other advantages, the present disclosure may address one or more of these needs.

(20) As used herein, the term “proximal,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve closest to the heart when the heart valve is implanted in a patient, whereas the term “distal,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve farthest from the heart when the heart valve is implanted in a patient. When used in connection with devices for delivering a prosthetic heart valve or other medical device into a patient, the terms “trailing” and “leading” are to be taken as relative to the user of the delivery devices. “Trailing” is to be understood as relatively close to the user, and “leading” is to be understood as relatively farther away from the user.

(21) The sealing portions of the present disclosure may be used in connection with collapsible prosthetic heart valves. FIG. 1 shows one such collapsible stent-supported prosthetic heart valve 100 including a stent 102 and a valve assembly 104 as is known in the art. Prosthetic heart valve 100 is designed to replace a native tricuspid valve of a patient, such as a native aortic valve. It should be noted that while the disclosures herein relate predominantly to prosthetic aortic valves having a stent with a shape as illustrated in FIG. 1, the valve could be a bicuspid valve, such as the mitral valve, and the stent could have different shapes, such as a flared or conical annulus section, a less-bulbous aortic section, and the like, and a differently shaped transition section.

(22) Prosthetic heart valve 100 will be described in more detail with reference to FIG. 1. Prosthetic



heart valve **100** includes expandable stent **102** which may be formed from biocompatible materials that are capable of self-expansion, such as, for example, shape memory alloys, such as the nickel-titanium alloy known as “Nitinol” or other suitable metals or polymers. Stent **102** extends from proximal or annulus end **130** to distal or aortic end **132**, and includes annulus section **140** adjacent proximal end **130**, transition section **141** and aortic section **142** adjacent distal end **132**. Annulus section **140** may have a relatively small cross-section in the expanded configuration, while aortic section **142** may have a relatively large cross-section in the expanded configuration. Preferably, annulus section **140** is in the form of a cylinder having a substantially constant diameter along its length. Transition section **141** may taper outwardly from annulus section **140** to aortic section **142**. Each of the sections of stent **102** includes a plurality of struts **160** forming cells **162** connected to one another in one or more annular rows around the stent. For example, as shown in FIG. 1, annulus section **140** may have two annular rows of complete cells **162** and aortic section **142** and transition section **141** may each have one or more annular rows of partial cells **162**. Cells **162** in aortic section **142** may be larger than cells **162** in annulus section **140**. The larger cells in aortic section **142** better enable prosthetic valve **100** to be positioned in the native valve annulus without the stent structure interfering with blood flow to the coronary arteries.

(23) Stent **102** may include one or more retaining elements **168** at distal end **132** thereof, retaining elements **168** being sized and shaped to cooperate with female retaining structures (not shown) provided on the deployment device. The engagement of retaining elements **168** with the female retaining structures on the deployment device helps maintain prosthetic heart valve **100** in assembled relationship with the deployment device, minimizes longitudinal movement of the prosthetic heart valve relative to the deployment device during unsheathing or resheathing procedures, and helps prevent rotation of the prosthetic heart valve relative to the deployment device as the deployment device is advanced to the target location and the heart valve deployed.

(24) Prosthetic heart valve **100** includes valve assembly **104** preferably secured to stent **102** in annulus section **140**. Valve assembly **104** includes cuff **176** and a plurality of leaflets **178** which collectively function as a one-way valve by coapting with one another. As a prosthetic aortic valve, valve **100** has three leaflets **178**. However, it will be appreciated that other prosthetic heart valves with which the sealing portions of the present disclosure may be used may have a greater or lesser number of leaflets.

(25) Although cuff **176** is shown in FIG. 1 as being disposed on the luminal or inner surface of annulus section **140**, it is contemplated that cuff **176** may be disposed on the abluminal or outer surface of annulus section **140** or may cover all or part of either or both of the luminal and abluminal surfaces. Both cuff **176** and leaflets **178** may be wholly or partly formed of any suitable biological material or polymer such as, for example, Polyethylene terephthalate (PET), ultra-high-molecular-weight polyethylene (UHMWPE), or polytetrafluoroethylene (PTFE).

(26) Leaflets **178** may be attached along their belly portions to cells **162** of stent **102**, with the commissure between adjacent leaflets **178** attached to commissure features **166**. As can be seen in FIG. 1, each commissure feature **166** may lie at the intersection of four cells **162**, two of the cells being adjacent one another in the same annular row, and the other two cells being in different annular rows and lying in end-to-end relationship. Preferably, commissure features **166** are positioned entirely within annulus section **140** or at the juncture of annulus section **140** and transition section **141**. Commissure features **166** may include one or more eyelets which facilitate the suturing of the leaflet commissure to stent **102**.

(27) Prosthetic heart valve **100** may be used to replace a native aortic valve, a surgical heart valve or a heart valve that has undergone a surgical procedure. Prosthetic heart valve **100** may be delivered to the desired site (e.g., near the native aortic annulus) using any suitable delivery device. During delivery, prosthetic heart valve **100** is disposed inside the delivery device in the collapsed configuration. The delivery device may be introduced into a patient using a transfemoral, transaortic, transsubclavian, transapical, transseptal or any other percutaneous approach. Once the

delivery device has reached the target site, the user may deploy prosthetic heart valve **100**. Upon deployment, prosthetic heart valve **100** expands so that annulus section **140** is in secure engagement within the native aortic annulus. When prosthetic heart valve **100** is properly positioned inside the heart, it works as a one-way valve, allowing blood to flow from the left ventricle of the heart to the aorta, and preventing blood from flowing in the opposite direction.

(28) FIG. 2 is a highly schematic cross-sectional illustration of prosthetic heart valve **100** disposed within native valve annulus **250**. As seen in the figure, valve assembly **104** has a substantially circular cross-section which is disposed within the non-circular native valve annulus **250**. At certain locations around the perimeter of heart valve **100**, gaps **200** form between heart valve **100** and native valve annulus **250**. Blood flowing through these gaps and past valve assembly **104** of prosthetic heart valve **100** can cause regurgitation and other inefficiencies which reduce cardiac performance. Such improper fitment may be due to suboptimal native valve annulus geometry due, for example, to calcification of native valve annulus **250** or to unresected native leaflets.

(29) FIG. 3A illustrates one embodiment of heart valve **300** intended to fill the irregularities between the heart valve and native valve annulus **250** shown in FIG. 2. Heart valve **300** extends between proximal end **302** and distal end **304**, and may generally include stent **306** and valve assembly **308** having a plurality of leaflets **310** and cuff **312**. Heart valve **300** may be formed of any of the materials and in any of the configurations described above with reference to FIG. 1.

(30) Stent **306** may include a plurality of struts **320**. Certain struts **320** may terminate in retaining elements **321** at distal end **304**. Struts **320** may come together to form cells **322** connected to one another in one or more annular rows around the stent. Connected to struts **320** are a plurality of runners **330**, which are additional struts that bow or bulge out radially when stent **306** is expanded, as will be described in greater detail with reference to FIGS. 3B and 3C.

(31) In order to better appreciate the attachment and placement of runners **330**, stent **306** is shown in FIG. 3B in its collapsed configuration. For the sake of clarity, valve assembly **308** is not shown in this figure. In the collapsed configuration of stent **306**, each of cells **322** is also collapsed. Stent **306** extends from proximal or annulus end **302** of heart valve **300** to distal or aortic end **304**, and includes annulus section **340** adjacent proximal end **302**, aortic section **342** adjacent distal end **304**, and transition section **341** between annulus section **340** and aortic section **342**. Commissure features **345** may be positioned entirely within annulus section **340** or at the juncture of annulus section **340** and transition section **341** as shown.

(32) One or more cells **322** may include runners **330**. An enlarged partial side view of cell **322** including a runner **330** is shown in FIG. 3C. Four struts **320a**, **320b**, **320c**, **320d** may join to form cell **322**, each strut being attached to two adjacent struts. In the collapsed configuration of stent **306**, cell **322** may be stadium-shaped as shown. In the expanded configuration of stent **306**, cell **322** may shorten in the length direction of stent **306** between proximal end **302** and distal end **304**, and struts **320** may generally form a diamond shape (FIG. 3D).

(33) Runners **330** may extend from first attachment end **335a** where struts **320a** and **320c** meet to second attachment end **335b** where struts **320b** and **320d** meet, and may be affixed to stent **306** by welding, adhesive, or any other suitable technique known in the art. Moreover, instead of being separately formed and affixed to stent **306** at attachment ends **335a**, **335b**, runners **330** may be integrally formed with stent **306**, such as by laser cutting both stent **306** and runners **330** from the same tube. Additionally, runners **330** may be formed of a shape memory material such as those described above for forming stent **102** of FIG. 1, and may have a substantially linear configuration in the collapsed configuration of heart valve **300** (FIG. 3C) and a curved or bowed configuration in the expanded configuration of heart valve **300** (FIG. 3D).

(34) In the collapsed configuration, runner **330** may bisect cell **322** into first portion **360a** and second portion **360b**. As the length of cell **322** shortens in the expanded configuration of heart valve **300**, the unchanged length of runner **330** causes the runner to bow or deflect outwardly of the curved surface defined by struts **320a**, **320b**, **320c**, **320d**. Stent **306** may also be heat set such that

struts **320** and runner **330** return to a predetermined shape in the fully expanded configuration (e.g., when no external forces are applied thereto). When cuff **312** (FIG. **3A**) is coupled to the abluminal surface of annulus section **340** of stent **306**, the cuff is substantially tubular when runners **330** are not bowed outwardly. When runners **330** bow outwardly on the expansion of heart valve **300**, they form protuberances in cuff **312** to help seal heart valve **300** within the native valve annulus.

(35) A method of delivering and implanting heart valve **300** will now be described with reference to FIGS. **4A-D**. A delivery system **400** may be used to deliver and deploy heart valve **300** in native valve annulus **250**, and may generally include sheath **410**, shaft **420**, atraumatic tip **430** and hub **440**. Sheath **410** may be slidable relative to shaft **420**. Heart valve **300**, including stent **306**, valve assembly **308** and runners **330**, may be disposed within sheath **410** about shaft **420** (FIG. **4A**). Hub **440** may be coupled to shaft **420** and configured to mate with retaining elements **321** of heart valve **300**. Runners **330** of heart valve **300** may be disposed in the linear configuration of FIG. **3C**, substantially parallel to sheath **410**, during delivery. Specifically, though runners **330** are configured to return to their curved configuration, they may be kept substantially linear by being constrained within sheath **410**. By doing so, heart valve **300** may be delivered to the native valve annulus using delivery system **400** without increasing the radius of sheath **410**, avoiding the need to increase the crimp profile of the heart valve within delivery system **400**. A large delivery system may be incapable of being passed through the patient's vasculature, while a delivery system having a heart valve with a smaller crimp profile may be easier to navigate through a patient's body and may also reduce the length of the implantation procedure. In the example shown in FIGS. **4A-D**, delivery system **400** is delivered from the aorta toward the left ventricle as indicated by arrow **S1**. If heart valve **300** or delivery system **400** includes echogenic materials, such materials may be used to guide delivery system **400** to the appropriate position using the assistance of three-dimensional echocardiography to visualize heart valve **300** within the patient. Alternative visualization techniques known in the art are also contemplated herein.

(36) When delivery system **400** has reached the proper location (e.g., atraumatic tip **430** is just past native valve annulus **250**), atraumatic tip **430** may be advanced slightly in the direction of arrow **S1** toward the left ventricle by pushing shaft **420** toward atraumatic tip **430** while holding sheath **410** in place, which serves to decouple atraumatic tip **430** from sheath **410** (FIG. **4B**). Sheath **410** may then be retracted in the direction of arrow **S2** toward the aorta. With sheath **410** slightly retracted, heart valve **300** begins to emerge from the sheath. As sheath **410** is further retracted in the direction of arrow **S2**, more of heart valve **300** is exposed until annulus section **340** is fully exposed and runners **330** become bowed (FIG. **4C**). Thus, sheath **410** may be retracted until heart valve **300** is free to self-expand within native valve annulus **250**. While heart valve **300** is partially deployed (e.g., a portion of heart valve **300** is outside sheath **410**, but heart valve **300** is not fully detached from delivery system **400**), if it appears that heart valve **300** needs to be recaptured and redeployed due to, for example, improper positioning or orientation, sheath **410** may be slid over shaft **420** in the direction of arrow **S1** to recapture heart valve **300** within sheath **410**. During recapture, sheath **410** may push against bowed runners **330** to straighten them to the linear configuration shown in FIG. **3C**. This process may be repeated until heart valve **300** is properly positioned and deployed within native valve annulus **250**.

(37) After sheath **410** has been fully retracted to expose heart valve **300**, runners **330**, now in their bowed or curved configuration, push cuff **312** outwardly against native valve annulus **250** and occlude gaps **200** between heart valve **300** and native valve annulus **250**, thereby reducing or eliminating the amount of blood that passes around heart valve **300** through gaps **200** (FIG. **4D**). Retaining elements **321** of heart valve **300** may decouple from hub **440** as heart valve **300** fully expands, atraumatic tip **430** may be retracted through heart valve **300** in the direction of arrow **S2** and delivery system **400** may be removed from the patient.

(38) Several variations of runners are described with reference to FIGS. **5A-10B**. In each variation, a collapsed cell is shown along with a schematic of the cell in the expanded configuration. As used

herein the terms expanded and collapsed may refer to the configurations of a cell, a stent, a heart valve and a valve assembly interchangeably.

(39) FIG. 5A shows cell **522** of a stent having four struts **520a**, **520b**, **520c**, **520d**, each strut being attached to two adjacent struts. In the collapsed configuration of the stent, cell **522** may be stadium-shaped as shown (FIG. 5A). Runner **530** may extend between two attachment ends **535a**, **535b**. Specifically, runner **530** may be joined to third strut **520c** at first attachment end **535a** and to diagonally-opposed second strut **520b** at second attachment end **535b**. Runner **530** may diagonally divide cell **522** substantially equally into first portion **560a** and second portion **560b** in the collapsed configuration. When cell **522** is placed in the expanded configuration (FIG. 5B), the cell may form a substantially diamond shape, with runner **530** stretching diagonally from strut second **520b** to third strut **520c** across the cell. Because attachment ends **535a**, **535b** are closer to one another in the expanded configuration than in the collapsed configuration, runner **530** may bow outwardly to form a protuberance.

(40) FIG. 6A shows cell **622** of a stent having four struts **620a**, **620b**, **620c**, **620d**, each strut being attached to two adjacent struts. Runner **630** may extend between two attachment ends **635a**, **635b** located at junctions of two struts (e.g., attachment end **635a** is at the junction of struts **620a** and **620c**, while attachment end **635b** is at the junction of struts **620b** and **620d**) and bisect cell **622** into substantially equal first portion **660a** and second portion **660b** in the collapsed configuration. Runner **630** may be tapered as shown, having a larger width at attachment ends **635a**, **635b** than at its middle. A tapered runner **630** may provide added flexibility and may be easier to heat set so that it readily returns to the bowed configuration when cell **622** is expanded, as shown in FIG. 6B.

(41) FIG. 7A shows cell **722** of a stent having four struts **720a**, **720b**, **720c**, **720d**, each strut being attached to two adjacent struts. Unlike cell **322**, runner **730** is joined to cell **722** at a single attachment end **735a** only and is able to deflect at free end **735b** opposite attachment end **735a**. It will be understood that the single point of attachment may be disposed at any of struts **720a**, **720b**, **720c**, **720d** or at the intersection of any two struts (e.g., at the intersection of second strut **720b** and fourth strut **720d** as shown, or at the opposite end at the intersection of first strut **720a** and third strut **720c**, or at the intersection of first strut **720a** and second strut **720b**, or third strut **720c** and fourth strut **720d**). In the collapsed configuration, runner **730** fills a portion of cell **722** so that a U-shaped cutout **760a** is formed within cell **722**. In the expanded configuration (FIG. 7B), runner **730** is capable of bowing radially outwardly to provide paravalvular sealing.

(42) FIG. 8A shows cell **822** of a stent having four struts **820a**, **820b**, **820c**, **820d**, each strut being attached to two adjacent struts. Cell **822** includes a pair of runners **830a**, **830b** that are substantially parallel to one another in the collapsed configuration. First runner **830a** is coupled to first strut **820a** at first attachment end **835a** and to second strut **820b** at second attachment end **835b**, while second runner **830b** is coupled to third strut **820c** at third attachment end **835c** and to fourth strut **820d** at fourth attachment end **835d**. Moreover, first and second runners **830a**, **830b** are coupled to one another at midpoint **845**, which keeps runners **830a**, **830b** close together along a midline of the diamond shaped cell **822** in the expanded configuration (FIG. 8B). Twin runners **830a**, **830b** may provide a larger support surface over which a cuff may be stretched to better seal a heart valve within a native valve annulus.

(43) FIG. 9A shows another variation having twin runners. Cell **922** includes four struts **920a**, **920b**, **920c**, **920d**, each strut being attached to two adjacent struts. Cell **922** includes a pair of runners **930a**, **930b** that are substantially parallel to one another in a collapsed configuration. First runner **930a** is coupled to the junction of first strut **920a** and third strut **920c** at first attachment end **935a** and to the junction of second strut **920b** and third strut **920d** at second attachment end **935b**, while second runner **930b** is coupled to the same two junctions at third attachment end **935c** and fourth attachment end **935d**. Runners **930a**, **930b** are not joined to one another except for having attachment ends near one another. When cell **922** expands, runners **930a**, **930b** bow outwardly and separate to provide scaffolding upon which a cuff may be stretched (FIG. 9B). Instead of the cuff

being stretched out over a single bowed runner or dual bowed runners attached at a midpoint, stretching the cuff over twin bowed runners that are spaced apart from one another provides a greater support area. Thus, this separation of bowed runners **930a**, **930b** may provide a more uniform protuberance for better sealing of a heart valve within the native valve annulus.

(44) Another variation, shown in FIG. **10A**, includes cell **1022** having four struts **1020a**, **1020b**, **1020c**, **1020d**, each strut being attached to two adjacent struts. Runner **1030** is formed as a U-shaped nested strut in the collapsed configuration, being attached to two adjacent struts **1020a**, **1020c** at attachment ends **1035a**, **1035b**, respectively. In the expanded configuration, runner **1030** bows radially outwardly to form a protuberance (FIG. **10B**). Because bowed runner **1030** is attached to adjacent struts **1020a**, **1020c** in the same half of cell **1022** and stretches between two attachment points at about the same longitudinal position, in the expanded configuration, bowed runner **1030** extends laterally across cell **1022**.

(45) Additionally, runners may be provided at a variety of locations on a stent. For example, in FIG. **11A** heart valve **1100A** extends between proximal end **1102** and distal end **1104**, and may generally include stent **1106** and valve assembly **1108** having a plurality of leaflets **1110** and cuff **1112**. Heart valve **1100A** may be formed of any of the materials and in any of the configurations described above with reference to FIG. **1**.

(46) Stent **1106** may include a plurality of struts **1120**, which may come together to form cells such as cell **1122A** connected to one another in one or more annular rows around the stent. Connected to struts **1120** are a plurality of runners **1130A**, which are additional struts that bow or bulge out radially when stent **1106** is expanded, as will be described in greater detail with reference to FIGS. **11B** and **11C**. As shown in FIG. **11A**, runners **1130A** are attached to the third full row of cells **1122A** from proximal end **1102** so that at least a portion of each runner is disposed radially outward of leaflets **1110**. In a second variation, shown in FIG. **11B**, heart valve **1100B** includes runners **1130B** attached to the first full row of cells **1122B** from proximal end **1102**. In yet another variation, shown in FIG. **11C**, heart valve **1100C** includes runners **1130C** attached to the bottom-most struts **1120C** so that they extend proximally of the proximal end **1102** of stent **1106**. It will be understood that the longitudinal position of runners **1130** may be varied anywhere within the annulus section and/or transition section. Additionally, multiple rows of runners may be disposed on stent **1106**. Moreover, each cell in an annular row of cells need not include a runner. Thus, there may be more runners in one annular row of cells than in another annular row of cells.

(47) FIG. **12** is a highly schematic cross-sectional view showing heart valve **1200** having stent **1202**, valve assembly **1204** including leaflets **1208** and a cuff **1222**, and bowed runners **1230** supporting portions of cuff **1222**. As seen in FIG. **12**, bowed runners **1230** extend radially outward from stent **1202** to press cuff **1222** into the gaps between heart valve **1200** and native valve annulus **250**. Cuff **1222** may be capable of promoting tissue growth between heart valve **1200** and native valve annulus **250**. For example, cuff **1222** may be innately capable of promoting tissue growth and/or may be treated with a biological or chemical agent to promote tissue growth, further enabling it to seal the heart valve within the native valve annulus. When runners **1230** are functioning properly, heart valve **1200** will be adequately sealed within native valve annulus **250** so that blood flows through leaflets **1208** of valve assembly **1204**, and so that blood flow through any gaps formed between heart valve **1200** and native valve annulus **250** is limited or reduced.

(48) The preceding embodiments have illustrated a simplified arrangement in which a cuff is disposed on the abluminal surface of a stent and attached to runners such that the expansion of the runners pushes the cuff outwardly toward walls of the native valve annulus. Other configurations of the cuff are also possible as illustrated below.

(49) FIG. **13** is a schematic developed view of a portion of heart valve **1300** including stent **1306** with cuff **1312** attached to same. For the sake of clarity, leaflets of the valve assembly are not shown. Stent **1306** includes a plurality of struts **1320** attached together to form diamond-shaped cells **1322** as has been described above. Three commissure features **1325a**, **1325b**, **1325c** are also

shown attached to struts **1320**. Runners **1330** are formed in each of cells **1322** in row **R1** located directly below commissure features **1325**.

(50) Cuff **1312** is disposed on the luminal surface of stent **1306** (i.e., FIG. **13** is a schematic illustration of the exterior of heart valve **1300**). Cuff **1312** may be formed of a polymer, a fabric or tissue, such as bovine, porcine, ovine, equine, kangaroo, PTFE, UHMWPE, PET, Dacron, PVA, Polyurethane, silicone or combinations thereof. Cuff **1312** includes diamond-shaped flaps **1350** for folding over certain cells **1322** in row **R1** that include runners **1330**. Each flap **1350** may be attached to cuff **1312** or formed integrally with cuff **1312** and may be folded at fold line **F1** from the luminal surface of stent **1306** to the abluminal surface of stent **1306** to cover runner **1330** as shown. A corresponding diamond-shaped suture pattern **P1** may attach a perimeter of flap **1350** to struts **1320** forming row **R1** of cells **1322** so that the bowing of runner **1330** pushes flap **1350** radially outward.

(51) Panels **1352** formed of the same or different material than cuff **1322** may be coupled to all cells or to certain cells **1322** where folding is difficult. Specifically, panels **1352** may be coupled to cuff **1312** and/or struts **1320** of cells **1322** of row **R1** located directly below commissure features **1325**, while flaps **1350** may be provided for all other cells **1322** in row **R1**. Panels **1352** may be formed from segments of material that are initially not attached to cuff **1312**, and may be attached to cells **1322** under commissure features **1325** using a diamond pattern of sutures similar to that of pattern **P1**. For the sake of clarity, only one panel **1352** is shown under the left-most commissure feature **1325a** and no panels are shown under commissure features **1325b**, **1325c** so that the shape of cuff **1312** may be appreciated. It will be understood, however, that panels **1352** may be coupled to cells **1322** located under commissure features **1325b** and **1325c**.

(52) FIG. **14** is a schematic developed view of a portion of heart valve **1400** including stent **1406** and cuff **1412** attached to the stent. Stent **1406** includes struts **1420** forming cells **1422**, commissure features **1425**, and runners **1430**. Heart valve **1400** includes first flaps **1450a** that are similar to flaps **1350** of FIG. **13** described above. Heart valve **1400** differs from heart valve **1300** in that second flaps **1450b** are also formed under commissure features **1425**. Specifically, instead of panels that initially are separate from and later are attached to the cuff, cuff **1412** includes second flaps **1450b** that fold 180 degrees at fold line **F2** and then attach to cell **1422** under commissure feature **1425** to form two layers of cuff sandwiching runner **1430**. For the sake of clarity, only one second flap **1450b** is shown under commissure feature **1425a**. Thus, cuff **1412** may be integrally formed from a single piece of material, disposed on the luminal surface of stent **1406** and include flaps on the abluminal surface to cover runners **1430**.

(53) FIG. **15** is a similar developed view of a portion of heart valve **1500** including stent **1506** with cuff **1512** attached thereto. Stent **1506** includes struts **1520** forming cells **1522**, commissure features **1525**, and runners **1530**. Various methods of folding flaps over runners **1530** are shown. In each example, flap **1550** is folded to a position over cell **1522** indicated by a diamond in broken lines. For example, flap **1550a** is capable of folding diagonally downward to position **1550a'**, while flaps **1550b** fold diagonally upward toward position **1550b'**. Flap **1550c** may fold straight down to position **1550c'** and flap **1550d** may fold diagonally downward to position **1550d'**.

(54) FIGS. **16A** and **16B** illustrate yet another example of cuff **1612** for covering a plurality of runners **1630** using flaps **1650**. FIG. **16A** illustrates the preassembled state in which cuff **1612** includes three flaps **1650a**, **1650b** and **1650c** that will be horizontally folded over runners **1630** and main portion **1660** of the cuff that covers six cells **1622**. Cuff **1612** may be formed of a single piece of material or may be formed as a composite cuff with multiple portions. For example, in FIG. **16A**, first flap **1650a** and second flap **1650b** may be formed of a first piece of material, main portion **1660** may be formed of a second piece of material and third flap **1650c** may be formed of a third piece of material. All three pieces may be sutured or otherwise coupled together to form a triple composite cuff.

(55) As seen in the assembled state (FIG. **16B**), first flap **1650a** and second flap **1650b** have been

folded horizontally so that first flap **1650a** extends to position **1650a'** over first runner **1630a**, and second flap **1650b** extends to position **1650b'** over second runner **1630b**. Flap **1650c** has been folded horizontally (in a direction opposite to first flap **1650a** and second flap **1650b**) to position **1650c'** over third runner **1630c**. Flaps **1650** may then be sutured to their respective cells around their perimeters

(56) In FIG. **17**, heart valve **1700** includes stent **1706** and cuff **1712**, as well as runners **1730** in row **R1** of cells **1722**. Instead of folding flaps as described above, individual panels **1750** are sutured around their perimeter to each cell **1722** in row **R1** as shown. Panels **1750** may be formed of the same material as cuff **1712**, or from a different material. Because runners **1730** bow outwardly, panels **1750** may be slightly larger than the underlying cells **1722**, such as first panel **1750a**, so as to not impede the outward bowing. Additionally, panels **1750** may be formed with a specific fiber orientation, compliance, thickness or the like to allow for the bowing. Pockets formed between cuff **1712** and panels **1750** may be filled with a liquid, a gel, a powder or other media to help support the outward bulging of the panels and thereby help mitigate paravalvular leakage. One example of the filler media may be a solution of polyvinyl alcohol (PVA). As cuff **1712** and panels **1750** contact blood upon the implantation of prosthetic heart valve **1700**, the filler media may swell in size, increasing the size and specifically the diameter of the pockets between the cuff and the panels. The enlarged pockets thus fill the gaps between the native valve annulus and the prosthetic heart valve, minimizing or preventing paravalvular leakage.

(57) FIG. **18** is a schematic developed view of a portion of heart valve **1800** including stent **1806** and cuff **1812** attached to the stent. Stent **1806** includes struts **1820** forming cells **1822**, commissure features **1825**, and runners **1830**. In FIG. **15**, runners **1330** were formed in each of cells **1322** in row **R1** located directly below commissure features **1325**. In the instant example, runners **1830** are provided in each of the cells in row **R2** located in the second full row below commissure features **1825**. Flaps **1850a** and **1850b** fold vertically downward to positions **1850a'** and **1850b'**, respectively, sandwiching certain runners **1830** between cuff **1812** and the flaps. Other runners **1830** may be covered with individual panels (not shown) as described above with reference to FIG. **17**.

(58) FIG. **19** illustrates another variation in which heart valve **1900** includes flaps **1950a-f** which are arranged to fold over runners **1930** in row **R2**. Flaps **1950a-f** may be unitarily formed with cuff **1912** and cut at certain edges depending on their intended position to create individual diamond-shaped flaps. Flaps **1950a-f** may be folded either diagonally or vertically downward to positions **1950a'-f'**, respectively. In another variation of folding flaps over the runners in second row **R2**, FIG. **20** illustrates heart valve **2000** having runners **2030** and flaps **2050a-f** that are foldable directly upward to the abluminal surface onto positions **2050a'-f'** as shown. In a further variation of the FIG. **20** embodiment, individual panels as shown in FIG. **17** may be disposed over each of runners **2030** in second row **R2**. Such panels may be enlarged or chosen based on specific properties as discussed above with reference to FIG. **17**.

(59) In FIG. **21**, heart valve **2100** includes stent **2106** and attached cuff **2112**. Stent **2106** includes a plurality of cells **2122** and runners **2130** in second row **R2** of the cells. Cuff **2112** is attached to the luminal surface of stent **2106**. Instead of individual panels or folded flaps, heart valve **2100** includes a unitary sheet of material **2150** coupled to the abluminal surface of stent **2106**. Sheet **2150** may be sized to extend over all cells **2122** in second row **R2** and may be attached to cuff **2112** using suture pattern **s21**. Suture pattern **S21** extends around the perimeter of sheet **2150** with the exception of edges **E1**, which may be left unattached to cuff **2112**. Blood flowing in a retrograde direction when heart valve **2100** is implanted in a patient may enter into a pocket **P21** formed between sheet **2150** and cuff **2112** at edges **E21** and causes the pocket to expand. In one variation of this embodiment, cuff **2112** and sheet **2150** may be unitarily formed such that a single piece of material is folded from the luminal surface over the bottom of the stent and forms a second layer on the abluminal surface of the stent. In one example, such a fold or transition may be made distal to

the runners.

(60) FIG. 22A is a schematic developed view of heart valve **2200** including stent **2206** and attached cuff **2212A**. Stent **2206** includes struts **2220** forming cells **2222**, commissure features **2225**, and runners **2230**. Three rows of cells, R1, R2, R3 are shown below commissure feature **2225**. In this example, runners **2230** are disposed in row R3 of incomplete cells defined at the proximal end of stent **2206**, and are attached to the bottom-most struts **2220** of stent **2206** in a manner similar to the stent described in connection with the embodiment of FIG. 11C. Individual panels **2250** may be sutured over each portion of a cell having a runner. In a slight variation, shown in FIG. 22B, cuff **2212B** may be disposed on the luminal surface of a stent (not shown) and include a plurality of triangular flaps **2260** that may be folded upwardly at fold line F22 and extend over the runners on the abluminal surface of the stent. Thus, flaps **2260** may take the place of individual panels **2250** from FIG. 22A.

(61) In FIG. 23, heart valve **2300** includes stent **2306** and attached cuff **2312**. Stent **2306** includes a plurality of cells **2322** arranged in three rows, R1, R2, R3 below commissure feature **2325**. Runners **2330** are disposed in row R3 of incomplete cells **2322** at the proximal end of the stent. Cuff **2312** is attached to the luminal surface of stent **2306**. Heart valve **2300** further includes a unitary sheet of material **2350** coupled to the abluminal surface of stent **2306** so as to overlie runners **2330**. Sheet **2350** may be sized to extend over all incomplete cells **2322** in row R3 and may be attached to cuff **2312** using a stitch pattern S23 around the perimeter of sheet **2350**. Thus, cuff **2312** and sheet **2350** may form a continuous pocket P23, which functions similarly to individual pockets P21 of FIG. 21 described above.

(62) FIG. 24 illustrates heart valve **2400** including stent **2406** and attached cuff **2412**. Stent **2406** includes a plurality of cells **2422** arranged in three rows, R1, R2, R3 below commissure feature **2425**. Runners **2430** are disposed in third row R3 of incomplete cells **2422** at the proximal end of the stent. Instead of a unitary sheet of material or individual panels, cuff **2412** includes upper portion **2450** and lower portion **2452** and fold line F24 separating the two portions. Fold line F24 is disposed along the tops of the incomplete cells in third row R3. Lower portion **2452** is divided into individual, substantially rectangular segments **2454**. Upper portion **2450** is attached to the luminal surface of stent **2406**. At fold line F24, rectangular segments **2454** may be folded from the luminal surface to the abluminal surface of stent **2406**, each segment passing through the center of cell **2412**. Segments **2454** of lower portion **2452** may then be stitched to adjacent segments and to runners **2430**. To aid in securing segments **2454** of lower portion **2452**, each runner **2430** may include a horseshoe **2460** at its free end and segments **2454** may be sutured to horseshoes **2460**, which may prevent lower portion **2452** from riding up runner **2430** during assembly or use. In some variations, eyelets or other similar features may be disposed at the free ends of the runners. Thus, a portion of cuff **2412** is disposed on the luminal surface of stent **2406** and a second portion of cuff **2412** is disposed over runners **2430** and on the abluminal surface of stent **2406**.

(63) While the devices disclosed herein have been described for use in connection with heart valve stents having a particular shape, the stent could have different shapes, such as a flared or conical annulus section, a less-bulbous aortic section, and the like, as well as a differently shaped transition section. Additionally, though the runners and cuffs have been described in connection with expandable transcatheter aortic valve replacement, they may also be used in connection with other expandable cardiac valves, as well as with surgical valves, sutureless valves and other devices in which it is desirable to create a seal between the periphery of the device and the adjacent body tissue.

(64) Moreover, although the disclosures herein have been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present disclosure. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present claims.



(65) In some embodiments, a prosthetic heart valve for replacing a native valve includes a stent extending between a proximal end and a distal end and including a plurality of struts forming cells, the stent having a collapsed condition and an expanded condition. At least one runner is coupled to a cell, the at least one runner being configured to transition from a first configuration to a second configuration when the stent moves from the collapsed condition to the expanded condition, the at least one runner projecting radially outwardly from the cell in the second configuration. A valve assembly is disposed within the stent, the valve assembly including a plurality of leaflets, a cuff at least partially disposed on a luminal surface of the stent and a covering material disposed on an abluminal surface of the stent and covering the at least one runner in the second configuration.

(66) In some examples, the at least one runner is substantially linear in the first configuration, and/or the at least one runner is bowed in the second configuration, and/or the cuff includes at least one flap configured to fold from the luminal surface to the abluminal surface of the stent to form the covering material, and/or the flap folds over a selected strut from the distal end toward the proximal end of the stent, and/or the flap folds over a selected strut from the proximal end toward the distal end of the stent, and/or the flap folds diagonally over a selected strut, and/or the at least one runner comprises multiple runners and the covering material includes a plurality of individual panels, each of the plurality of individual panels being disposed over one of the multiple runners, and/or the covering material includes at least one diamond-shaped panel disposed over at least one cell, and at least one flap integrally formed with the cuff and configured to fold from the luminal surface to the abluminal surface of the stent, and/or the stent further includes commissure features and the at least one runner coupled to selected cells in a first row of the cells proximal of the commissure features, and/or the stent further includes commissure features and the at least one runner coupled to selected cells in a second row of the cells proximal of the commissure features, and/or a plurality of runners coupled to proximal-most struts of the stent and extending proximally of the proximal end of the stent, and/or each of the plurality of runners extends from a first end joined to a strut to a free end, and/or each free end has a horseshoe shape, the covering material being sutured to the horseshoe shape, and/or the covering material includes a sheet of material circumferentially disposed on the abluminal surface of the stent so as to extend over a row of cells.

(67) In some embodiments, a prosthetic heart valve for replacing a native valve includes a stent extending between a proximal end and a distal end and including a plurality of struts forming cells and a plurality of runners, the stent having a collapsed condition and an expanded condition, the struts defining a first diameter and the runners defining a second diameter, the second diameter being greater than the first diameter. A valve assembly is disposed within the stent, the valve assembly including a plurality of leaflets and a cuff at least partially disposed on a luminal surface of the stent and partially disposed on an abluminal surface of the stent to cover the runner.

(68) In some examples, the stent includes a plurality of commissure features and the cuff includes an upper portion and a lower portion, the upper portion being disposed on the luminal surface of the stent and coupled to the commissure features and the lower portion being disposed on the abluminal surface of the stent so as to cover the plurality of runners, and/or the lower portion is divided into individual segments capable of passing through the cells from the luminal surface to the abluminal surface, and/or the cuff transitions from the luminal surface to the abluminal surface distal to the plurality of runners, and/or each of the individual segments is coupled to an adjacent segment and a runner.

(69) It will be appreciated that the various dependent claims and the features set forth therein can be combined in different ways than presented in the initial claims. It will also be appreciated that the features described in connection with individual embodiments may be shared with others of the described embodiments.

## Claims

1. A prosthetic heart valve for replacing a native valve, comprising: a stent extending between a proximal end and a distal end along a longitudinal axis and including a plurality of struts forming cells; a plurality of leaflets disposed within the stent; a cuff disposed on a surface of the stent and defining an inflow of a valve assembly and an outflow of a valve assembly, a portion of the cuff being folded along a folding axis different from the longitudinal axis to define an inner layer and an outer layer, wherein a portion of the cuff is folded in a diagonal direction with respect to the longitudinal axis.
2. The prosthetic heart valve of claim 1, further comprising at least one runner coupled to a respective cell of the stent and sandwiched between the inner layer of the cuff and the outer layer of the cuff, the at least one runner being configured to transition from a first configuration to a second configuration when the stent moves from a collapsed condition to an expanded condition.
3. The prosthetic heart valve of claim 2, wherein the at least one runner has first and second ends coupled to opposing vertices of one of the cells.
4. The prosthetic heart valve of claim 2, wherein the at least one runner is substantially linear in the first configuration and is parallel to the longitudinal axis of the stent.
5. The prosthetic heart valve of claim 2, wherein the at least one runner is bowed or curved in the second configuration.
6. The prosthetic heart valve of claim 2, wherein the at least one runner projects radially outwardly from one of the cells in the second configuration.
7. The prosthetic heart valve of claim 2, wherein the at least one runner is disposed radially outward of the leaflets.
8. The prosthetic heart valve of claim 2, wherein the stent has a length direction between the proximal end and the distal end, and the at least one runner extends substantially in the length direction in the second configuration.
9. The prosthetic heart valve of claim 2, wherein the at least one runner extends substantially in a circumferential direction of the stent in the second configuration.
10. The prosthetic heart valve of claim 1, wherein a portion of the cuff is folded in an orthogonal direction with respect to the longitudinal axis.
11. A prosthetic heart valve for replacing a native valve, comprising: a stent extending between a proximal end and a distal end along a longitudinal axis and including a plurality of struts forming cells; a plurality of leaflets disposed within the stent; a cuff disposed on a surface of the stent and defining an inflow of a valve assembly and an outflow of a valve assembly, a portion of the cuff being folded along a folding axis different from the longitudinal axis to define an inner layer and an outer layer, the outer layer being moveable relative to the inner layer to create pockets, wherein the cuff includes multiple flaps that are foldable in different directions.
12. The prosthetic heart valve of claim 11, further comprising at least one runner coupled to a respective cell of the stent, the at least one runner being configured to transition from a first configuration to a second configuration when the stent moves from a collapsed condition to an expanded condition, and wherein each of the cells include a lower vertex adjacent the proximal end, an upper vertex adjacent the distal end, and two side vertices disposed longitudinally between the upper vertex and the lower vertex.
13. The prosthetic heart valve of claim 12, wherein the at least one runner has first and second ends coupled to opposing vertices of one of the cells.
14. The prosthetic heart valve of claim 12, wherein the at least one runner is substantially linear in the first configuration and is parallel to the longitudinal axis of the stent.
15. The prosthetic heart valve of claim 12, wherein the at least one runner is bowed or curved in the second configuration.
16. The prosthetic heart valve of claim 12, wherein the at least one runner projects radially outwardly from the one of the cells in the second configuration.

17. The prosthetic heart valve of claim 11, wherein a portion of the cuff is folded in an orthogonal direction with respect to the longitudinal axis.

18. The prosthetic heart valve of claim 11, wherein a portion of the cuff is folded in a diagonal direction with respect to the longitudinal axis.

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